Modelling the Episode-Oriented Medical Record according to Solon in openEHR: Patterns for Longitudinal, Problem- and Episode-Centred Care

Jean-Pierre Messerli[1]

[1]Independent Medical Informatics Researcher, Switzerland
Correspondence: Jean-Pierre Messerli (jean-pierre.messerli@jpm-consulting.ch)

Abstract

Background

Electronic health records (EHRs) have become the foundation of modern healthcare, yet organising longitudinal clinical information remains a persistent challenge affecting continuity of care, clinical reasoning, auditability and secondary use. Most systems continue to structure documentation around source systems, documents, encounters or isolated problem lists, providing only limited support for representing the complete course of individual health problems across multiple contacts, healthcare professionals and care settings.

In 1967, Solon and colleagues introduced the concept of the episode-oriented medical record, proposing that healthcare documentation should be organised around episodes of care related to individual health problems rather than around documents or encounters [1]. Although this concept has substantially influenced longitudinal medical record design, no systematic implementation strategy has been described for representing Solon's methodology within a vendor-neutral openEHR environment.

openEHR provides a standards-based persistence framework built on a stable Reference Model, archetypes, templates, versioning, provenance and powerful query capabilities, while intentionally avoiding predefined application-specific organisational models [2,3]. This raises an important architectural question: how can an episode-oriented medical record be realised within openEHR while preserving semantic integrity, longitudinal queryability, provenance and interoperability?

Objective

This study develops and systematically evaluates openEHR-conformant implementation patterns for representing Solon's episode-oriented medical record. It compares their suitability for longitudinal, problem- and episode-centred clinical documentation and derives practical recommendations for semantic governance, provenance, queryability and vendor-neutral health information systems.

Methods

The study followed a design-science research methodology supported by a conceptual proof-of-concept implementation. Solon's methodology was analysed and translated into a conceptual information model comprising three complementary contextual dimensions: *care contacts* as the chronological documentation spine, *episodes of care* as the longitudinal clinical context for individual health problems, and *administrative encounters* as organisational periods for grouping and reporting. Together, these dimensions establish the conceptual framework of an episode-oriented medical record.

The model was subsequently mapped to existing openEHR structures without extending the specification. The mapping incorporated COMPOSITIONs, ENTRYs, FOLDERs, LINKs and archetyped CLUSTERs to represent longitudinal relationships while remaining fully compliant with the openEHR architecture [2,3].

Three implementation patterns were developed and systematically compared: (1) folder-based directory indexing, (2) link-based referencing between openEHR LOCATABLEs, and (3) a cluster-based relationship model using archetyped relationship attributes. A structured evaluation framework assessed openEHR conformance, semantic expressiveness, longitudinal persistence, AQL queryability, provenance,

governance effort, implementation complexity, clinical interpretability and suitability for multi-vendor and cross-repository environments.

The proposed approach was evaluated using a realistic longitudinal patient history comprising 132 clinical entries covering acute, chronic and resolved health problems. Clinical applicability was further assessed through eight clinician-centred user stories addressing chronological navigation, episode-oriented reconstruction, section-based documentation, diagnosis and problem lists, past medical history and alternative longitudinal views.

Results

The study proposes an openEHR-based conceptual architecture demonstrating how Solon's episode-oriented medical record can be represented entirely within existing openEHR specifications without modifying the Reference Model. The resulting architecture distinguishes three complementary contextual dimensions: the *clinical section*, defining the type of clinical information, the *care contact*, providing the event-level documentation context, and the *episode of care*, representing the longitudinal health-problem context. Together, these dimensions establish a coherent organisational framework for longitudinal clinical documentation.

Episodes of care are represented as governed openEHR COMPOSITIONs serving as the single source of truth for the longitudinal health-problem context. Care contacts remain explicit documentation events, while clinical statements are recorded within clinical COMPOSITIONs and ENTRYs. Consequently, every medical record entry is simultaneously anchored within its clinical section, care contact and episode of care, preserving provenance, traceability and semantic integrity.

Evaluation using the 132-entry longitudinal patient history and eight clinician-centred user stories confirmed that all required clinician-centred views – including chronological records, episode-centred records, diagnosis lists, problem lists and past medical history – can be generated consistently from the same underlying clinical information without duplication of clinical data.

The comparative evaluation demonstrated complementary strengths across the three implementation patterns. The *folder-based pattern* enables intuitive clinician navigation and longitudinal organisation but requires governance of folder lifecycle, naming and update policies. The *link-based pattern* supports flexible cross-composition and cross-entry relationships but depends on consistent semantic conventions and increases query and application complexity. The *cluster-based pattern* provides the strongest semantic governance by representing episode relationships and contextual attributes as explicit archetyped clinical content that is directly coded, queryable and reusable while concentrating modelling complexity within reusable templates and CLUSTERs.

No single implementation pattern satisfied all identified requirements. The recommended solution is therefore a *hybrid implementation* combining governed CLUSTERs as the authoritative semantic representation, a FOLDER-based index for intuitive patient-wide navigation, and selective LINKs for explicit cross-record relationships where additional referential semantics are required.

Within this approach, episode lists, diagnosis lists, problem lists and past medical history are derived clinical views rather than independently maintained data structures, preserving semantic integrity, provenance and auditability within a single underlying clinical information model, consistent with contextual problem-list approaches [4].

Discussion

The findings demonstrate that episode-oriented clinical documentation can be realised within the existing openEHR architecture without modifications to the Reference Model, confirming that longitudinal clinical context can be represented using established openEHR concepts. Rather than prescribing a single implementation strategy, the study provides systematic implementation guidance enabling

solution architects and implementers to balance semantic governance, clinical usability, queryability and implementation complexity according to their deployment context.

More broadly, the study demonstrates that clinically meaningful organisational models can be realised through archetypes, templates and governance principles rather than through changes to the underlying technical architecture. The proposed hybrid approach illustrates how openEHR can support longitudinal, problem- and episode-centred care while preserving interoperability, provenance and auditability.

Conclusions

This study presents one of the first systematic implementation frameworks for representing Solon's episode-oriented medical record in openEHR. By integrating care contacts, episodes of care, clinical sections and administrative encounters into a coherent conceptual framework and comparatively evaluating three complementary implementation patterns, it establishes a practical and vendor-neutral approach to longitudinal clinical documentation.

The proposed hybrid implementation provides a reusable architectural blueprint for future openEHR-based longitudinal health records, supports derived clinical views from a single semantic information model, and offers practical guidance for semantic governance, implementation projects and future standardisation activities.

Keywords

openEHR; episode of care; episode-oriented medical record; Solon; longitudinal health records; clinical information modelling; semantic interoperability; vendor-neutral architecture; electronic health record

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References

1. Solon JA, Feeney JJ, Jones SH, Rigg RD, Sheps CG. Delineating episodes of medical care. Am J Public Health Nations Health. 1967;57(3):401–408.

2. Beale T, Heard S. openEHR Architecture Overview [Internet]. London: openEHR Foundation; 2024 [cited 2025 Aug 18]. Available from: https://specifications.openehr.org/.

3. openEHR Specification Program. EHR Information Model (RM Release 1.1.0) [Internet]. London: openEHR Foundation; c2003–2025 [cited 2025 Sep 1]. Available from: https://specifications.openehr.org/releases/RM/latest/ehr.html.

4. Meredith J, McNicoll I, Whitehead N, Ademoye K. Defining the contextual problem list. In: Pape-Haugaard LB, et al., editors. Digital Personalized Health and Medicine. Amsterdam: IOS Press; 2020. p. 567–571. Studies in Health Technology and Informatics; 270. doi:10.3233/SHTI200224.

Figures

Figure 1. Conceptual data model of the episode-oriented medical record. The model separates contacts, episodes of care, medical record entries, clinical sections and derived diagnosis/problem lists. Each clinical entry is anchored in a care contact, assigned to an episode of care and classified by a clinical section, while terminology bindings support semantic interoperability.

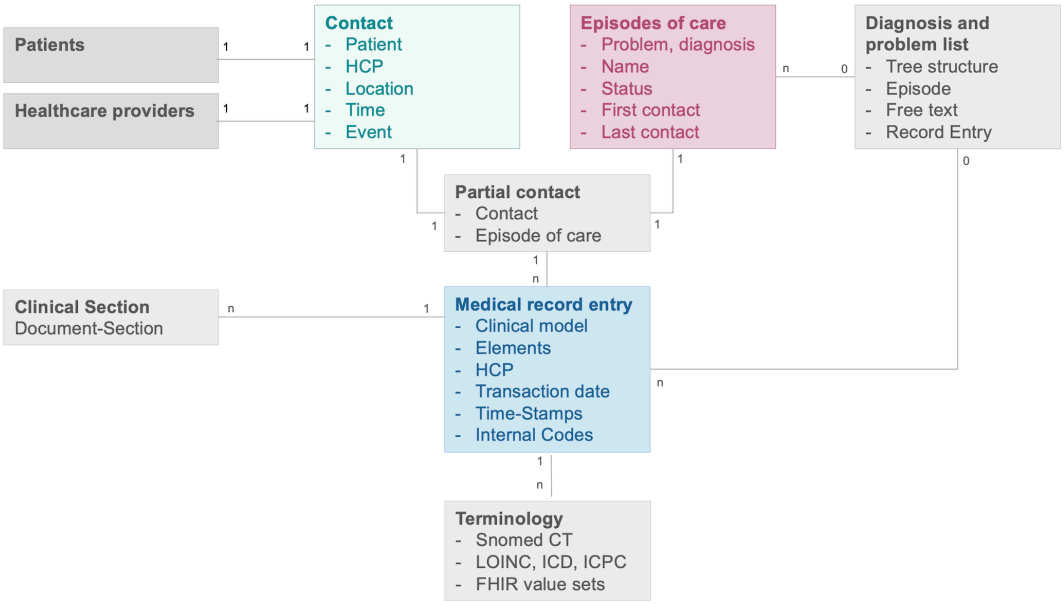


Figure 1. Conceptual data model of the episode-oriented medical record.

Figure 2. openEHR mapping of contacts, medical record entries and episodes of care. Administrative encounters may group several care contacts. Each contact may give rise to one or more clinical COMPOSITIONs containing archetyped ENTRYs. Relationship attributes at archetype or template level associate the documented clinical content with the relevant Episode-of-Care, preserving both event context and longitudinal problem context.

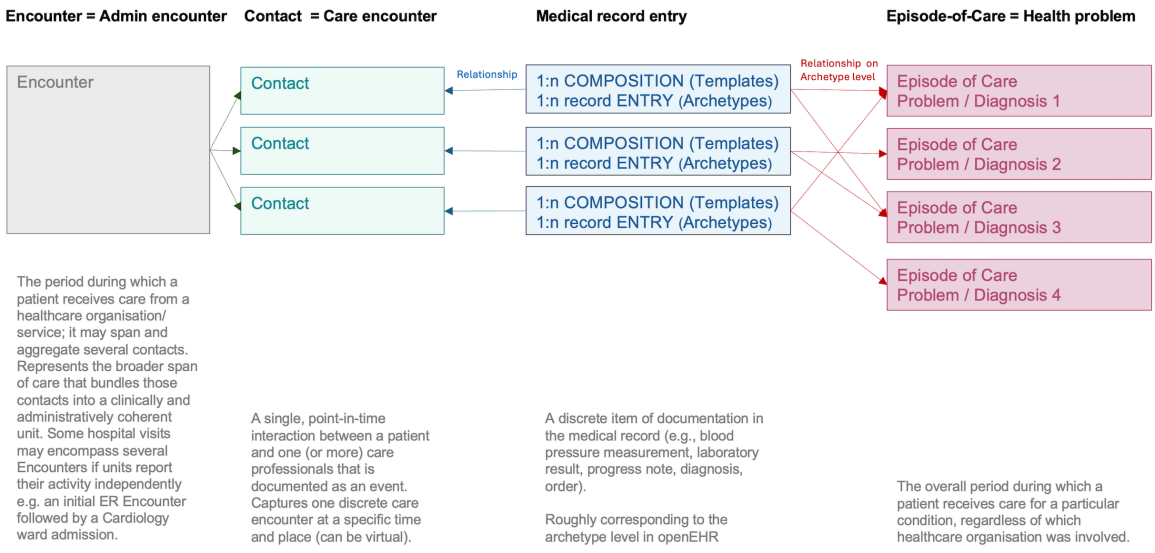


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